CME Clearing Overview

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Presentation Outline

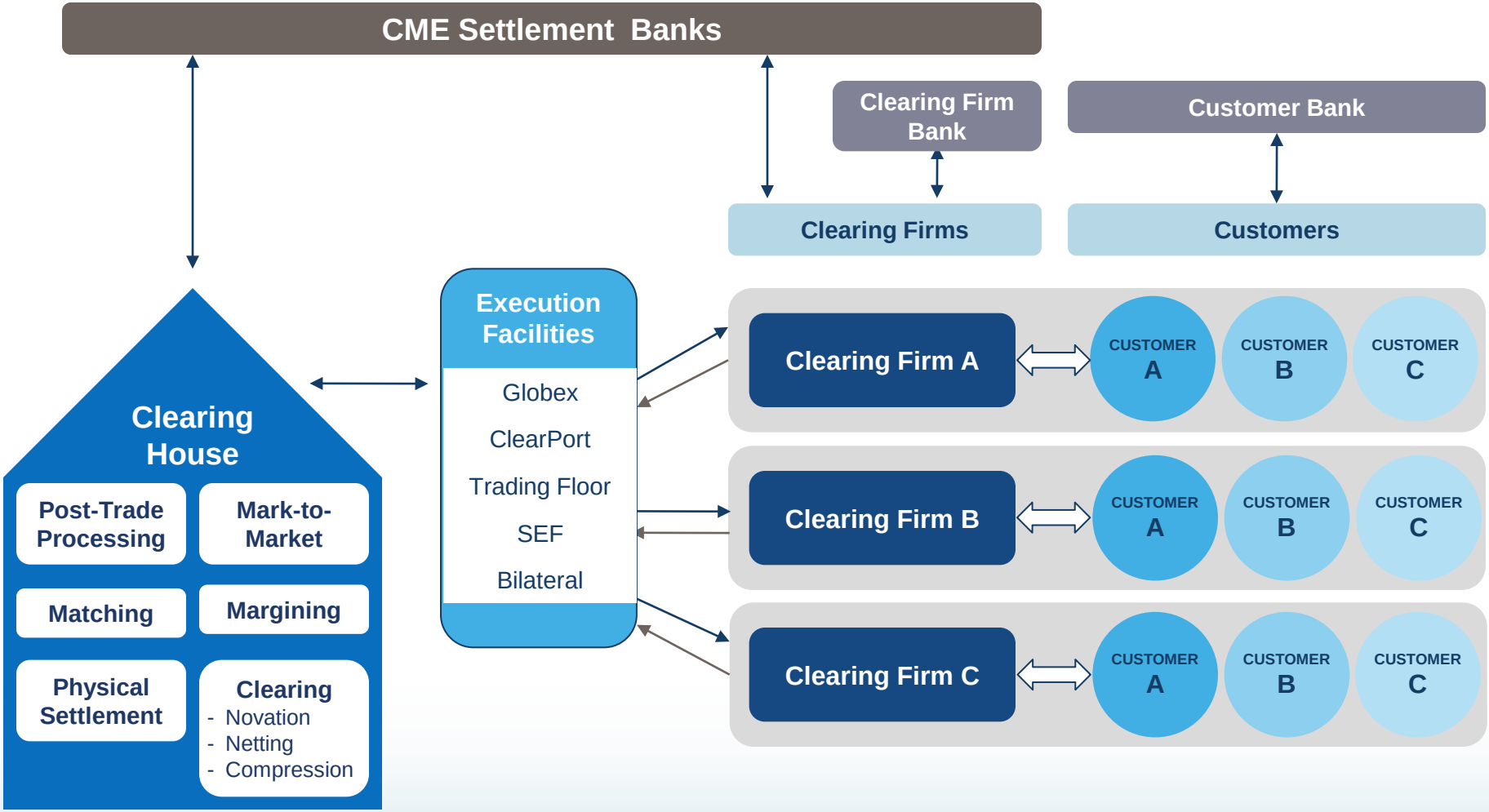
1. Clearing Market Structure
2. Risk Management Overview
3. Performance Bond (Margin) Coverage
4. CME Span® Margin Model
5. Settlement Overview
6. Collateral Protection and Segregation
7. Financial Safeguards

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CME Clearing Market Structure

Clearing Market Structure

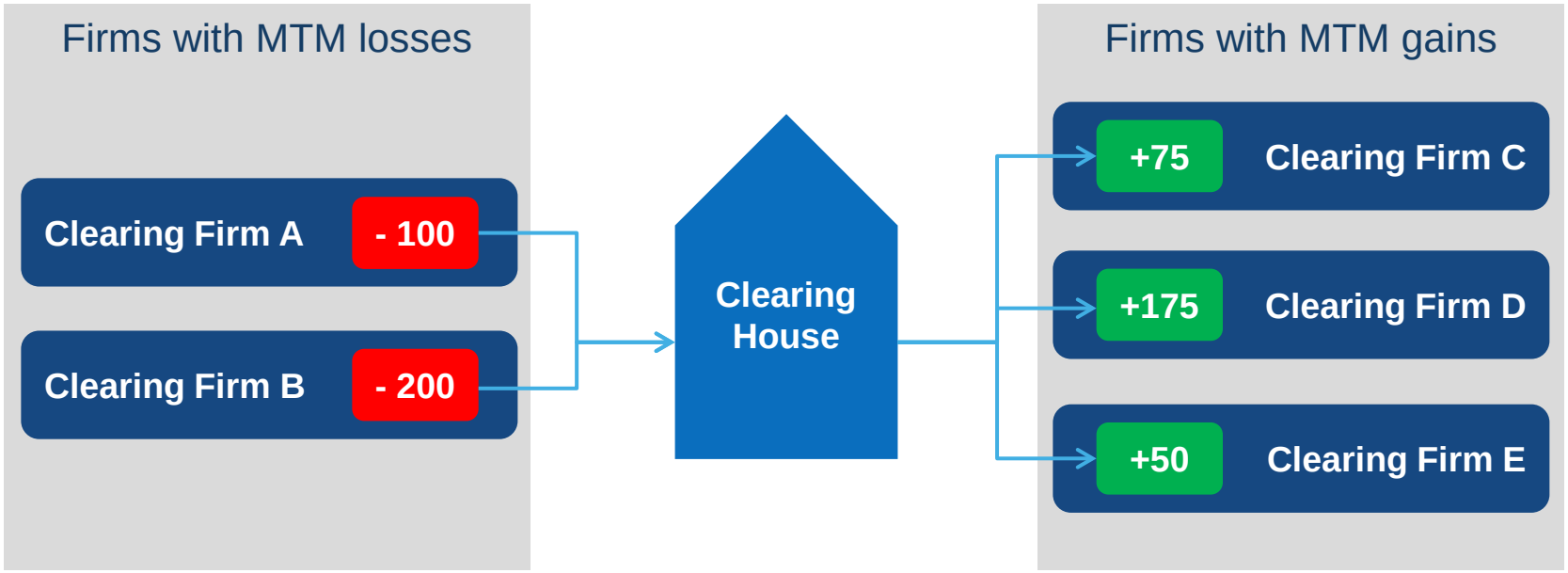


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Clearing Market Structure

Daily Mark-to-Market



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CME Risk Management

Risk Management Overview

CME's Risk management principles have allowed CME to achieve excellence in exchange traded space:

- Price transparency
- Anonymous matching
- Twice daily mark-to-market
- Clearing member oversight
- Customer funds segregation
- Over $8B Financial Safeguards*
- Industry-leading market regulation
- Strength of central counterparty

Exchange – Traded Derivatives
• Electronic execution via formal exchanges • Exchange rules and regulations to govern trading • Submitted to CME Clearing for central counterparty clearing

Over-The-Counter (OTC) Derivatives
• Voice, Broker, and electronic execution via traditional OTC methods and practice • "Standardized" but flexibility to specify • Underlying instrument • Single and multiple settlement payment dates

* Number is only for Exchange traded products. The OTC Financial Safeguards package is $500M plus assessments

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Risk Management Overview

Daily Risk Monitoring

Mark-to-market

- Real-time positions are marked to real-time prices and are available immediately for risk analysis to prevent the accumulation of debt

Market Oversight

- Market, operational, liquidity, and credit risk are continuously and effectively monitored on a daily basis
- Maintain Clearing House and Audit department staff in excess of 250 to implement daily processes and apply a variety of sophisticated tools. Monitoring is performed 24 hours a day, 6 days a week
- Overnight risk management activities include monitoring overnight price moves, Globex activity levels, mark-to-market accumulation of trading, and tolerance reviews

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Performance Bonds Coverage

CME SPAN® - Performance Bonds Coverage

- Performance bond levels and requirements are evaluated routinely using:
  - Back testing of coverage levels vs. actual underlying price volatility
  - Back testing of portfolio requirements vs. actual portfolio losses
- Performance bond requirements for all products cover at least a 1-day 99% confidence interval
- Volatilities are updated daily for all products
- At CME, we revise margin requirements to reflect any changes in market conditions, volatility spikes, or correlations between products
- Margins are determined through analysis of several quantitative methodologies

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CME SPAN® - Quantitative Models for Margins

Historical Percentiles

- Historical data is used to estimate the current volatility by choosing the percentile observation that margin should cover

Value at Risk (VaR)

- Evaluates potential losses, given a certain level of confidence, that may be encountered over a specified time period due to adverse price movements

Normal Mixtures

– Uses mixing of distributions to improve the estimation of events outside the reach of a normal distribution

Extreme Value Theory (EVT)

– Extreme market events are used to calculate a probability distribution

– Focuses on the tail behavior of extreme returns, a good indication of market risk under extreme events like the ’87 crash

Exponentially Weighted Moving Average (EWMA)

– The relevance of old observations decreases exponentially as more recent observations are taken into consideration. As time progresses, old observations are replaced with new ones

Implied Volatility Levels

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SPAN® Margining

CME SPAN® - Standard Portfolio Analysis of Risk

- Developed by Chicago Mercantile Exchange Inc. to effectively assess risk on an overall portfolio basis
- SPAN is a market simulation based Value At Risk system which has been reviewed and validated by market regulators and participants world wide
- SPAN is the official Performance Bond mechanism of 54 exchanges and clearing organizations world-wide, making it the global standard for portfolio margining
- SPAN's risk based margin requirements allows for effective margin coverage while preserving efficient use of capital
- SPAN assesses risk for a wide variety of financial instruments including: futures, options, physicals, equities, or any combination

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CME SPAN® - Objectives

- SPAN assesses the risk of a portfolio, by calculating the maximum likely loss that could be suffered by the portfolio based on parameters set by the margin-setting authority, usually an exchange or clearing organization
- The core of SPAN risk analysis is to simulate potential market moves and calculate the profit or loss on individual contracts given the market moves
- Exchanges may determine any number of market scenarios to be included in the SPAN analysis
  - Most SPAN exchanges and clearing organizations use 16 scenarios

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CME SPAN® - Methodology

- SPAN groups together financial instruments with the same underlying for analysis
- For example, Futures on an Equity Index and Options on the Equity Index would be grouped together for analysis
- Each product is referred to as a Combined Commodity
- SPAN uses parameters set by the exchange or clearing organization to evaluate a portfolio with the following two step analysis:

  Step 1: Analyze the risk of each Combined Commodity in isolation from other Combined Commodities

  Step 2: Seek risk reducing offsets between Combined Commodities

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CME SPAN® - Parameter Files

- Exchanges & Clearing Organizations that license SPAN can publish Risk Parameter Files daily with updated information such as listed products, performance bond requirements, spreads, etc
- Risk Parameter Files can be produced several times a day
- CME currently produces 7 files a day
- Each SPAN Exchange or Clearing Organization is free to use its own criteria in setting margin parameters
- CME Clearing evaluates:
  - Short, medium, and longer term historical price moves
  - Option Implied Volatility
  - Macro-economic factors
  - Liquidity & concentration in the market
- The margin-setting authority has complete control over the margin parameters or degree of margin coverage

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CME SPAN® - Parameter Setting

SPAN allows for the following parameters to be set by the margin-setting authority:

Price Scan Range

- The range of potential price changes in which to analyze potential profits and losses

Volatility Scan Range

- The range of potential implied volatility changes for options on the underlying instrument

Intra-Commodity Spread Charge

- An amount that accounts for risk (basis risk) of calendar spreads or different expirations of the same products, which are not perfectly correlated

Inter-Commodity Spread Credit

- Margin credit for offsetting positions between correlated products

Short Option Minimum

- Minimum margin requirement for short option positions

Spot Charge

- A charge that covers the increased risk of positions in deliverable instruments near expiration

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CME SPAN® - Scan Risk

- SPAN Risk Arrays represent a contract's hypothetical gain/loss under a specific set of market conditions from a set point in time to a specific point in time in the future
- The risk array representing the maximum likely loss becomes the Scan Risk for the portfolio

Consider an example risk array for an S&P500 portfolio with two positions:

- Long 1 Sep 2012 SP Futures (price = 1100)
- Short 1 Sep 2012 SP 1000 Call Option (implied volatility = 28%)

Assume:

- Volatility Scan Range for SP500 = 7%
- Price Scan Range = $22,500 or 90 points
  - CVF for S&P500 is $250, $22,500/$250 = 90 points

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CME SPAN® - Scan Risk Example

Scenario	SP Underlying Price Move	Volatility Move	SP Future Gain/Loss	SP Option Gain/Loss	Portfolio Gain/Loss
1	UNCHANGED	UP	0	1,807	1807
2	UNCHANGED	DOWN	0	-1,838	-1,838
3	UP 33%	UP	-7,499	7,899	400
4	UP 33%	DOWN	-7,499	5,061	-2,438
5	DOWN 33%	UP	7,499	-3,836	3,663
6	DOWN 33%	DOWN	7,499	-8,260	-761
7	UP 67%	UP	-15,001	14,360	-641
8	UP 67%	DOWN	-15,001	12,253	-2,748
9	DOWN 67%	UP	15,001	-8,949	6,052
10	DOWN 67%	DOWN	15,001	-13,980	1,021
11	UP 100%	UP	-22,500	21,107	-1,393
12	UP 100%	DOWN	-22,500	19,604	-2,896
13	DOWN 100%	UP	22,500	-13,455	9,045
14	DOWN 100%	DOWN	22,500	-18,768	3,732
15	UP 300%	UNCHANGED	-22,275	21,288	-987
16	DOWN 300%	UNCHANGED	22,275	-9,160	13,115
				Largest Potential Loss = SPAN Risk	**13,115**

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CME SPAN® - Spread Types and Formation

Super Inter-Commodity Spread

- Allows Inter-Commodity Spreads to be evaluated before Intra-Commodity Spreads

Intra-Commodity Spread

- Evaluate the basis risk between contract periods with different expirations within the same product
- Spreads are prioritized by lowest charge

Inter-Commodity Spread

- Evaluate credit available for offsetting positions in related instruments
- Spreads are prioritized by greatest total savings

Inter-Exchange Spread Credit

- Allows spreads to be formed for portfolios containing products listed on multiple Exchanges, as defined by the Exchange
  - The formation of Inter-Exchange Spreads is similar to process of forming Inter-Commodity Spreads, however each Exchange can only provide a credit for its own products

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CME SPAN® - Short Option Minimum

- Deep out-of-the-money short options may show zero or minimal Scan Risk given the price & volatility moves in the 16 market scenarios
- However, in extreme events these options may move closer to-the-money or in-the-money, thereby generating potentially large losses
- To account for this potential exposure, a Short Option Minimum can be set for each product
- If the Scan Risk is lower than the Short Option Minimum, then the Short Option Minimum is charged
- The next slide shows an example of the Short Option Minimum using a deep out-of-the-money short put
  - Short 1 SP500 Sep 2012 Put @500 (underlying price is 1100)
  - Short Option Minimum on 1 SP500 is $225

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CME SPAN® - Short Option Minimum Example

Scan Risk is $88, however SOM is $225, so the requirement is $225.

Scenario	Underlying Price Move	Volatility Change	Gain/Loss
1	UNCHANGED	UP	$16
2	UNCHANGED	DOWN	-$10
3	UP 33%	UP	$8
4	UP 33%	DOWN	-$10
5	DOWN 33%	UP	$27
6	DOWN 33%	DOWN	-$9
7	UP 67%	UP	$3
8	UP 67%	DOWN	-$11
9	DOWN 67%	UP	$41
10	DOWN 67%	DOWN	-$7
11	UP 100%	UP	-$1
12	UP 100%	DOWN	-$11
13	DOWN 100%	UP	$62
14	DOWN 100%	DOWN	-$4
15	UP 300%	UNCHANGED	-$4
16	**DOWN 300%**	**UNCHANGED**	**$88**
		Scan Risk	$88

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CME SPAN® - Summary of SPAN Analysis

- **Scan Risk**: Evaluate the directional market risk
- **Intra-Commodity Spread Charge**: Evaluate the basis risk between contract periods with different expirations within the same product
- **Inter-Commodity Spread Credit**: Evaluate credit available for offsetting positions in related instruments
- **Delivery Add-On Charge**: Evaluate contract periods for increasing volatility during delivery
- **Short Option Minimum**: Evaluate short option positions for potential increased risk, using the greater of the Scan Risk or Short Option Minimum

SPAN Requirement for a Combined Commodity is the greater of:

- Scan Risk + Intra Commodity Spread Charge + Delivery Charge – Inter Commodity Spread Credit
- Short Option Minimum

- **The Total SPAN Requirement for a portfolio is the sum of the SPAN Requirement for all Combined Commodities**

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CME SPAN® - Net Option Value

- Mark-to-market of options is reflected in the Net Option Value component of SPAN
- The Total Performance Bond Requirement for a portfolio reflects the Total SPAN Requirement and the Net Option Value of the portfolio
- **The Net Option Value (NOV)**: equal to the Long Option Value minus the Short Option Value for the portfolio
- **Long Option Value (LOV)**: The total value of all the long options in the portfolio
- **Short Option Value (SOV)**: The total value of all the short options in the portfolio
- LOV *reduces* the overall Total Performance Bond Requirement
- SOV *increases* the overall Total Performance Bond Requirement

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CME SPAN® - Stress Testing

- SPAN can also perform stress testing functionality, allowing the user to analyze the profit/loss impact of specified market conditions
- Stress testing parameters include:
  - Price
  - Volatility
  - Time to Expiration
  - Interest Rate
  - Dividend Yield
- Stress tests may include one or more products and parameters may vary by product
- Stress test results include the impact on the Greeks of the portfolio including Delta, Gamma, Vega, Theta, and Rho
- Stress test impact can be viewed down to the contract level

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CME SPAN® - Family Of Software Products

- PC-SPAN
  - Single-user desktop application that offers margin calculation across multiple exchanges
  - Includes scripting language called ‘Span-it’, facilitating margin calculation once a day
- SPAN Risk Manager
  - Performs PC-Span margin calculation.
  - Provides stress testing capabilities including risk analytics, detailed position analysis, and “What-If” scenario analysis of potential market impact on profit/loss on portfolios
- SPAN Real-Time Component Interface (RTCI)
  - Com API that facilitates near real-time margin calculations & recalculation of requirements throughout the day using a trade/position feed
- SPAN Risk Manager Clearing (RM Clearing)
  - Institutional-level program used by exchanges, clearing organizations, service bureaus, and regulatory agencies
  - Employs all the functionalities of PC-SPAN, SPAN Risk Manager, & SPAN RTCI, offering real-time margining, risk array calculations, and production of SPAN risk parameter files
  - Provides an end-to-end SPAN solution
  - Written in Microsoft Visual C++, executes on Microsoft Windows platform

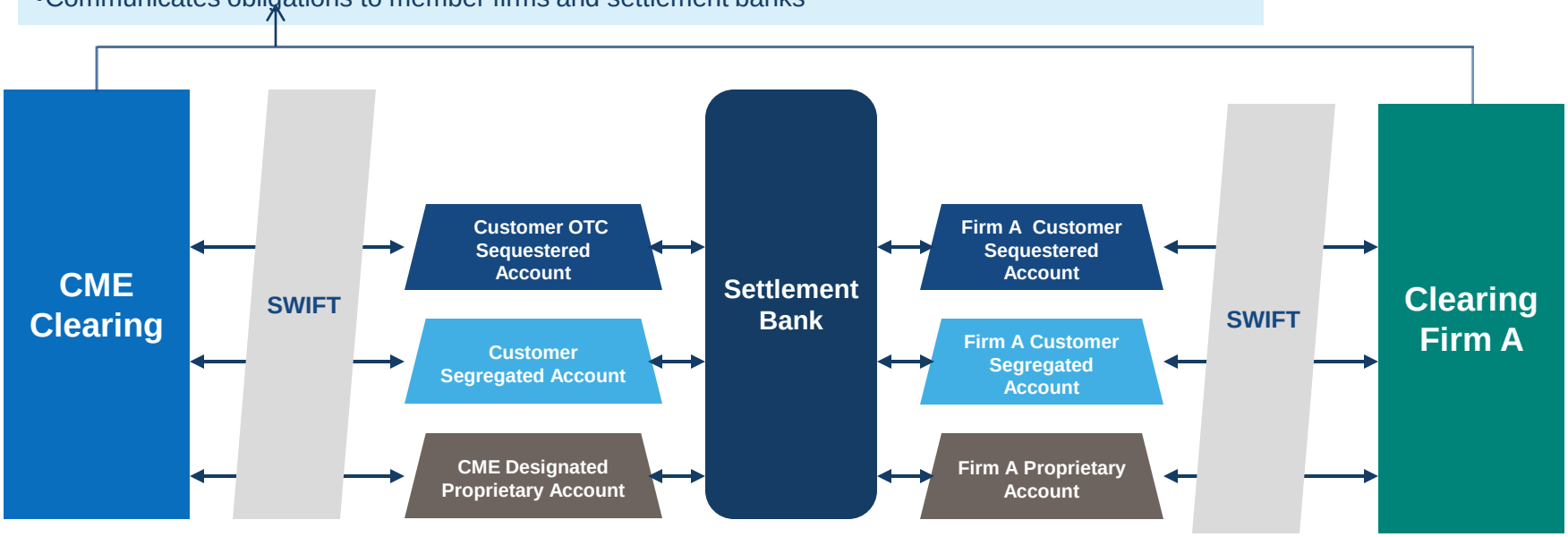
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Settlement Overview

Settlement Cash Flows

STEP 1
CME Clearing:
- Calculates portfolio level margin requirements
- Calculates mark-to-market banked in cash
- Communicates obligations to member firms and settlement banks

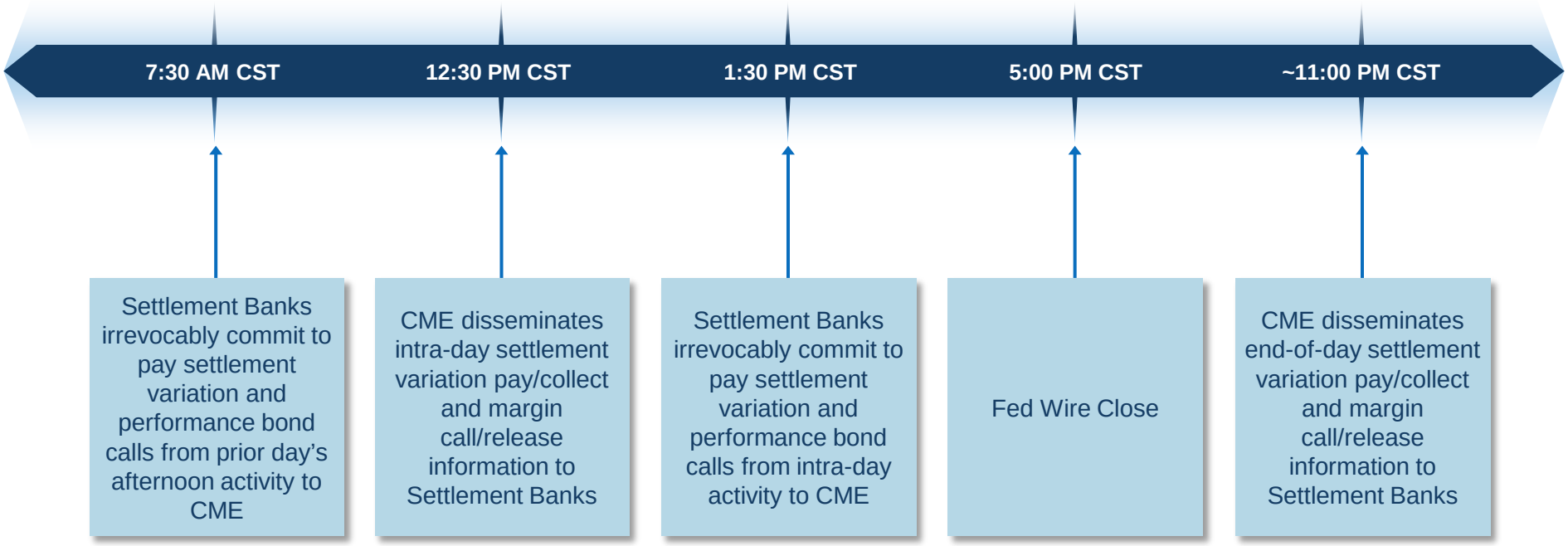


STEP 2
CME Clearing:
- Generates SWIFT messages with banking instructions
- Initiates auto collects/pays with Customer Segregated Account
- Initiates auto collects/pays with Proprietary Account
- Initiates auto collects/pays with Customer OTC Sequestered Account

CME Settlement Banks	
BMO Harris	Bank of New York
Bank of America	Fifth Third
JP Morgan Chase	Lakeside Bank
Brown Brothers Harriman	

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Daily Settlement Process



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Collateral Protection & Segregation Methodology

Acceptable Collateral

The Clearing House determines the acceptability of various collateral types and appropriate haircuts as well as category type. Firms can meet up to 40% of their margin requirements with category 2 assets and the lesser of 40% of their margin requirements or $3B with category 3 assets.

Summary of acceptable collateral[1]:

Category	Asset Class	Description
1	**Cash**	USD or up to 250mm of eligible FX to meet obligations in a different currency (per firm, per currency)
	U.S. Treasuries	TBILL, TBOND, TNOTE, TSTRIP
	IEF2	Specialized Collateral Program Money market mutual fund program
2	**U.S. Government Agencies & Select Mortgage Backed Securities**	• Eligible issuers include Fannie Mae, Freddie Mac, Federal Farm Credit Bank, Federal Home Loan Bank, and also include Ginnie Mae for MBS securities • Issue size must be greater than $1 billion for coupon bearing instruments (GSEOTH) and the only permissible derivative feature is floating rate • Mortgage backed securities must be backed by 15 or 30 year conventional loans only; fixed rate only
	TLGP	Temporary Liquidity Guarantee Program • Issue size greater than $1 billion • Issuer may not be affiliated with the clearing member, its parent or any of its affiliates • Must be delivered with Repo Tracking on DTC • $200 million cap per firm
	IEF5 Specialized Cash Program	Cash with interest on trust ledger of participating banks
	Letters of Credit	Letters of credit for performance bond
3	**Gold**	Physical gold • $500m cap per firm • Agreements are required
	Select Foreign Sovereign Debt	Select discount bills, notes, bonds • $1bn per firm
	Stock	Select stocks from S&P 500 index
	IEF3 & IEF4	Specialized Collateral Program Fee-based tri-party collateral management program for pledging corporate bonds

[1]Refer to CME Group website for most currently available data: http://www.cmegroup.com/clearing/financial-and-collateral-management/

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Customer Segregation

CME Clearing is dedicated to customer protection and segregation

- CFTC Regulations and Exchange Rules require all funds and collateral received from a customer for U.S. futures trading to be held separately from the clearing member's funds
- The purpose of customer segregation is to protect customer funds if the clearing member suffers significant losses or goes bankrupt
- ***All*** funds or collateral received from a customer trading on U.S. futures markets must be held in an account titled "customer segregated"
- The depository must acknowledge that all funds held in these accounts must be strictly for the use of customers and can not offset any funds owed to the institution by the clearing member
- Qualified depositories include banks, U.S. FCMs and registered derivatives clearing organizations

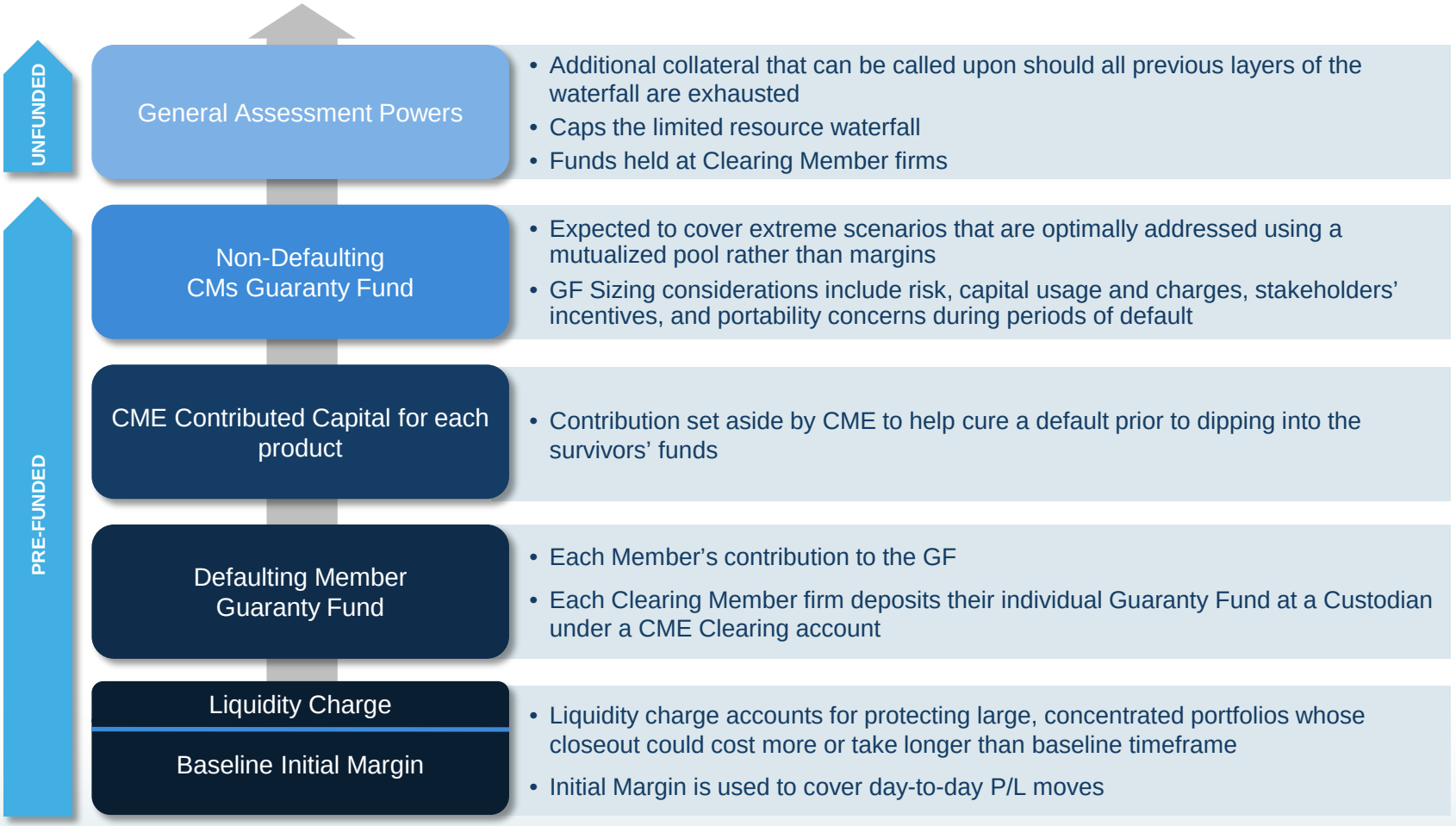
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Financial Safeguards

Financial Safeguards

Guaranty Fund

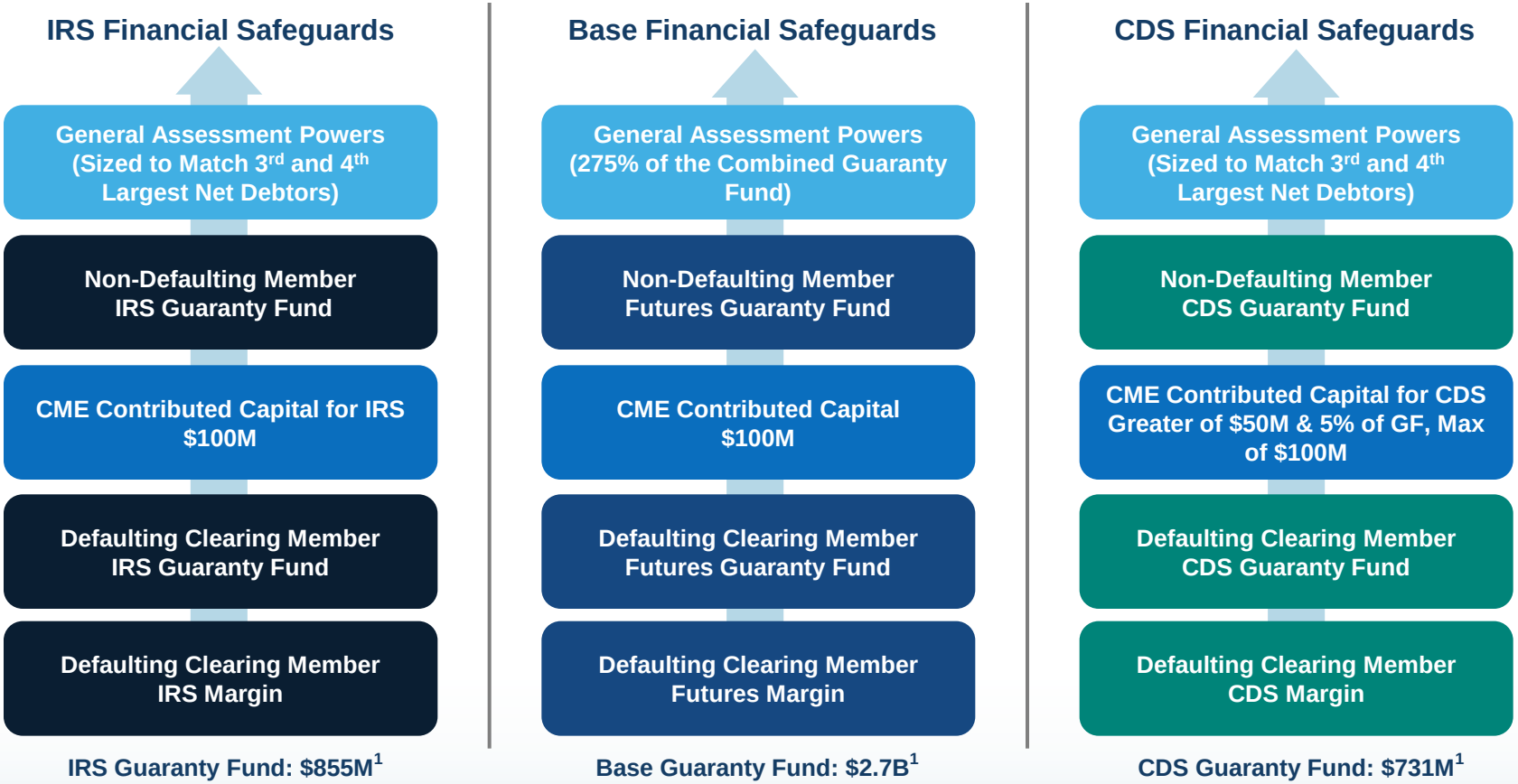


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Financial Safeguards

The Guaranty Fund is an integral part of the CME Clearing Financial Safeguard program



[1]Approximate requirements as of 6/29/2012

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Q & A

Futures trading is not suitable for all investors, and involves the risk of loss. Futures are a leveraged investment, and because only a percentage of a contract's value is required to trade, it is possible to lose more than the amount of money deposited for a futures position. Therefore, traders should only use funds that they can afford to lose without affecting their lifestyles. And only a portion of those funds should be devoted to any one trade because they cannot expect to profit on every trade.

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